On the Secrecy Performance of Pinching-Antenna Systems

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***Abstract*—Pinching-antenna systems have recently gained significant attention as a novel reconfigurable-antenna technology due to its exceptional capability of mitigating signal-propagation path loss. In this paper, we investigate the secrecy performance of a pinching-antenna system in the presence of an eavesdropper. In particular, we derive an approximate expression of the system's secrecy outage probability (SOP) with respect to the random locations of the legitimate user and eavesdropper and analyze its asymptotic behavior. Moreover, we derive a constant performance lower bound on the SOP of the considered system, i.e., $\frac{2\pi-1}{24}$, which is significantly lower than that of conventional fixed-position antenna systems, i.e., 0.5. Finally, simulation results are provided to validate the correctness of our analytical results.**

I. Introduction

WITH rapid proliferation of various wireless applications, such as Internet of Things (IoTs), telemedicine, high-quality video streaming, the upcoming sixth generation (6G) communication networks are poised to meet the versatile demand of higher capacity, lower latency, and enhanced reliability. However, signal-propagation path loss, a long-standing challenge in wireless communications, poses a critical hurdle to realizing the above vision. In this context, various reconfigurable antenna systems have been developed in recent years as a viable technology to break through this obstacle [1], [2], [3], [4]. In particular, pinching antennas utilize a long dielectric waveguide as signal transmission mediums and enable flexible antenna activation at any location along the waveguide by deploying small dielectric particles. This innovative antenna architecture reduce the signal-propagation distance between the antenna's radiating points and users, thereby offering great potential to mitigate the large-scale path loss [1], [5].

Motivated by such a promising benefit, many recent works have optimized the performance of pinching-antenna systems in various wireless communication scenarios. In [6], the authors investigated an uplink sum-rate maximization problem in pinching-antenna aided multiuser communication systems. In [7], the authors studied the performance optimization for downlink multiuser pinching-antenna systems. Besides, the authors of [8] proposed to apply pinching antennas for integrated sensing and communications (ISAC). Furthermore, exploring pinching antennas for empowering physical layer security was considered in [9], [10]. However, most of the existing works focus their attention on antenna-positioning optimization, while an in-depth theoretical analysis of pinching-antenna systems still remains largely unexplored.

To fill in this gap, this paper conducts performance analysis of a pinching antenna-enabled secure communication system. As shown in Fig. 1, we consider the downlink transmission from a pinching-antenna base station (BS) to a single-antenna legitimate user (Bob) in the presence of a single-antenna eavesdropper (Eve). Our objective is to analyze the system's secrecy outage probability (SOP) with respect to the random locations of Bob and Eve. First, by applying the Gaussian-Chebyshev quadrature, we obtain an analytical approximate expression of the SOP, followed by which its asymptotic behavior is given. Then, to gain more insights, we derive a constant lower bound on the SOP of our considered pinching-antenna system and compare with that of traditional fixed-position antenna (FPA) systems, showing that the former is significantly lower than the latter ($\frac{2\pi-1}{24} \approx 0.2201$ versus 0.5). Finally, numerical results are provided to validate the correctness of our analytical results. It is worth noting that a recent work in [11] has also analyzed the performance of a similar secure pinching-antenna system. However, it assumes identical signal-to-noise ratio (SNR) distributions for Bob and Eve. In contrast, this paper studies a more practical scenario where Bob and Eve are randomly distributed with different resulted SNR distributions and provides a deterministic performance bound.

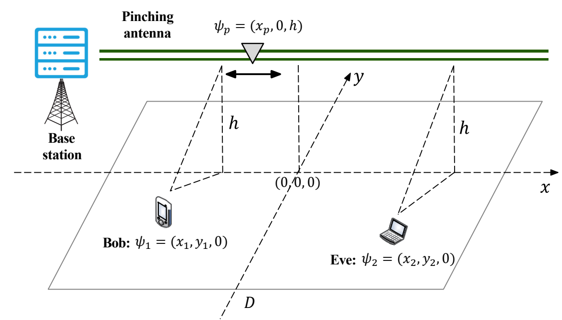

Fig. 1. Secure pinching-antenna system.

II. System Model

As illustrated in Fig. 1, we consider the downlink of a secure communication system, which consists of a BS, a legitimate user (Bob), and an eavesdropper (Eve) aiming to intercept the information intended for Bob. Both Bob and Eve are equipped with a single FPA while the BS employs a pinching antenna on a dielectric waveguide for enhancing the system's secrecy performance. The waveguide is assumed to be installed parallel to the x-axis at a height h. Hence, the position of the pinching antenna can be denoted by $\boldsymbol{\psi}_p = (x_p, 0, h)$. Besides, the positions of Bob and Eve are assumed to be uniformly distributed within the x-y plane with a size of $D \times D$

$$f_{\gamma_E}(z)=\begin{cases}\frac{\bar\gamma}{z^2}\left[\frac{\pi}{D^2}-\frac{2}{D^3}\sqrt{\frac{\bar\gamma}{z}-h^2}\right], & \frac{\bar\gamma}{h^2+\frac{D^2}{4}}\le z\le\frac{\bar\gamma}{h^2},\\ \frac{2\bar\gamma}{z^2D^2}\left[\arcsin\left(\frac{D}{2\sqrt{\frac{\bar\gamma}{z}-h^2}}\right)-\frac{1}{2}\right], & \frac{\bar\gamma}{h^2+D^2}\le z<\frac{\bar\gamma}{h^2+\frac{D^2}{4}},\\ \frac{2\bar\gamma}{z^2D^3}\left[D\left(\arcsin\left(\frac{D}{2\sqrt{\frac{\bar\gamma}{z}-h^2}}\right)-\arccos\left(\frac{D}{\sqrt{\frac{\bar\gamma}{z}-h^2}}\right)\right)-\left(\frac{D}{2}-\sqrt{\frac{\bar\gamma}{z}-h^2-D^2}\right)\right], & \frac{\bar\gamma}{h^2+\frac{5D^2}{4}}\le z<\frac{\bar\gamma}{h^2+D^2},\\ 0, & \text{otherwise.}\end{cases} \quad (5)$$

and are denoted by $\boldsymbol{\psi}_1=(x_1,y_1,0)$ and $\boldsymbol{\psi}_2=(x_2,y_2,0)$, respectively. In this paper, we assume Eve is passive, such that the BS only has Bob's channel state information (CSI). Thus, the pinching antenna is activated at the position that is closest to Bob, i.e., $\tilde{\boldsymbol{\psi}}_p=(x_1,0,h)$.

A. SNR at Bob

In this paper, we adopt the spherical wave model to characterize the channel between the antenna activation point and Bob. Thus, the received signal at Bob is expressed as[1]

$$r_B=\sqrt{P_s}\frac{\eta^{\frac{1}{2}}e^{-j\frac{2\pi}{\lambda}\|\tilde{\boldsymbol{\psi}}_p-\boldsymbol{\psi}_1\|}}{\|\tilde{\boldsymbol{\psi}}_p-\boldsymbol{\psi}_1\|}e^{-j\phi}s+n_B, \quad (1)$$

where $\eta=\frac{\lambda^2}{16\pi^2}$, with λ denoting the signal wavelength. P_s denotes the transmit power of the BS, s denotes the transmitted signal from the BS with a unit power, and n_B denotes the additive white Gaussian noise (AWGN) with zero mean and variance σ^2. Besides, $\phi=\frac{2\pi}{\lambda_g}\|\tilde{\boldsymbol{\psi}}_p-\boldsymbol{\psi}_0\|$ represents the phase shift incurred by the signal propagation inside the waveguide, where $\boldsymbol{\psi}_0=(-\frac{D}{2},0,h)$ denotes the feed-point position of the transmitted signal and $\lambda_g=\frac{\lambda}{n_{\text{eff}}}$ is the guided wavelength in the waveguide.

As a result, the SNR at Bob can be obtained as

$$\gamma_B=\frac{\eta P_s}{\|\tilde{\boldsymbol{\psi}}_p-\boldsymbol{\psi}_1\|^2\sigma^2}=\frac{\bar\gamma}{y_1^2+h^2}, \quad (2)$$

where $\bar\gamma=\frac{\eta P_s}{\sigma^2}$ denotes the effective transmit SNR. Based on [12, Proposition 2], the cumulative distribution function (CDF) of γ_B with respect to the random position of Bob is given by

$$F_{\gamma_B}(z)=\begin{cases}1, & z\ge\frac{\bar\gamma}{h^2},\\ 1-\frac{2}{D}\sqrt{\frac{\bar\gamma}{z}-h^2}, & \frac{\bar\gamma}{h^2+\frac{D^2}{4}}\le z\le\frac{\bar\gamma}{h^2},\\ 0, & z\le\frac{\bar\gamma}{h^2+\frac{D^2}{4}}.\end{cases} \quad (3)$$

B. SNR at Eve

Similarly, the SNR at Eve can be obtained as

$$\gamma_E=\frac{\eta P_s}{\|\tilde{\boldsymbol{\psi}}_p-\boldsymbol{\psi}_2\|^2\sigma^2}=\frac{\bar\gamma}{(x_1-x_2)^2+y_2^2+h^2}. \quad (4)$$

Since x_1, x_2, and y_2 are independent random variables with uniform distributions over $[-\frac{D}{2},\frac{D}{2}]$, the probability density function (PDF) of γ_E can be obtained as (5), shown at the top of this page. The details are provided in Appendix A.

[1] In this work, to facilitate our analysis and drive essential insights, we do not consider the inside-waveguide attenuation and assume continuous antenna-location activation. This enables a rigorous characterization of the performance limit of the considered secure pinching-antenna system.

C. Performance Metric

In this paper, we adopt the SOP as the performance metric of the considered secure pinching-antenna system. Specifically, we assume that the BS transmits signals to Bob at a constant target secrecy rate, R_{th}, and a secure transmission can only be guaranteed if R_{th} is lower than the system's instantaneous achievable secrecy rate. The SOP is, therefore, defined as

$$\mathcal{P}_{so}=\Pr\left(\log_2(1+\gamma_B)-\log_2(1+\gamma_E)\le R_{th}\right), \quad (6)$$

which characterizes the probability that a secure transmission cannot be realized.

III. Performance Analysis

In this section, we derive the analytical expression of the system's SOP as well as its asymptotic behavior and drive insights by comparing with conventional FPA systems.

A. SOP Analysis

According to (6), the SOP can be equivalently expressed as

$$\begin{aligned}\mathcal{P}_{so}&=\Pr\left(\frac{1+\gamma_B}{1+\gamma_E}\le C_{th}\right)\\&=\Pr\left(1+\gamma_B-C_{th}\le C_{th}\gamma_E\right),\end{aligned} \quad (7)$$

where $C_{th}=2^{R_{th}}$. By utilizing the CDF of γ_B and the PDF of γ_E in (3) and (5), the SOP can be further expanded as

$$\begin{aligned}\mathcal{P}_{so}&=\int_0^{+\infty}f_{\gamma_E}(t)\Pr\left(\gamma_B\le C_{th}t+C_{th}-1\right)dt\\&=\int_{\frac{\bar\gamma}{h^2+\frac{5D^2}{4}}}^{\frac{\bar\gamma}{h^2}}f_{\gamma_E}(t)F_{\gamma_B}\left(C_{th}t+C_{th}-1\right)dt.\end{aligned} \quad (8)$$

However, due to the cumbersome integration, it is still challenging to analyze the SOP based on (8). Therefore, we present the following theorem, which provides an approximate closed-form expression of the system's SOP.

Theorem 1. *For the considered secure pinching-antenna system, the SOP can be approximated by*

$$\begin{aligned}\mathcal{P}_{so}\approx\frac{\pi}{N}\sum_{n=1}^{N}&\sqrt{1-\cos^2\left(\frac{2n-1}{N}\pi\right)}\Gamma\left(\cos\left(\frac{2n-1}{N}\pi\right)\right)\\&\times\left(\frac{\bar\gamma}{2h^2}-\frac{\bar\gamma}{2h^2+\frac{5D^2}{2}}\right),\end{aligned} \quad (9)$$

where $\Gamma(x)=f_{\gamma_E}\left(\left(\frac{\bar\gamma}{2h^2}-\frac{\bar\gamma}{2h^2+\frac{5D^2}{2}}\right)x+\frac{\bar\gamma}{2h^2}+\frac{\bar\gamma}{2h^2+\frac{5D^2}{2}}\right)\cdot F_{\gamma_B}\left(C_{th}\left(\left(\frac{\bar\gamma}{2h^2}-\frac{\bar\gamma}{2h^2+\frac{5D^2}{2}}\right)x+\frac{\bar\gamma}{2h^2}+\frac{\bar\gamma}{2h^2+\frac{5D^2}{2}}\right)+C_{th}-1\right)$

and N is a tunable parameter to balance the complexity and accuracy.

Proof: First, let us define $u = \frac{2t-\left(\frac{\bar{\gamma}}{h^2}+\frac{\bar{\gamma}}{h^2+\frac{5D^2}{4}}\right)}{\frac{\bar{\gamma}}{h^2}-\frac{\bar{\gamma}}{h^2+\frac{5D^2}{4}}}$. Then, the integral interval in (8) can be transformed into $[-1, 1]$, i.e.,

$$\mathcal{P}_{so} = \int_{-1}^{1} f_{\gamma_E}\left(g(u)\right) F_{\gamma_B}\left(C_{th} g(u) + C_{th} - 1\right) \times \left(\frac{\bar{\gamma}}{2h^2} - \frac{\bar{\gamma}}{2h^2+\frac{5D^2}{2}}\right) du, \tag{10}$$

where $g(u) = \left(\frac{\bar{\gamma}}{2h^2} - \frac{\bar{\gamma}}{2h^2+\frac{5D^2}{2}}\right) u + \frac{\bar{\gamma}}{2h^2} + \frac{\bar{\gamma}}{2h^2+\frac{5D^2}{2}}$. Next, by applying N-point Gaussian-Chebyshev quadrature [13], we can derive the approximated SOP in (9). □

B. Asymptotic Behavior of SOP

To gain further insights, we analyze the asymptotic behavior of the system's SOP with respect to the transmit power of the BS. First, for a sufficiently large P_s, i.e., $P_s \to \infty$, we have $\bar{\gamma} = \frac{\eta P_s}{\sigma^2} \to \infty$. Therefore, the terms $\log_2(1+\gamma_B)$ and $\log_2(1+\gamma_B)$ in (6) can be approximated by $\log_2(1+\gamma_B) \to \log_2\left(\frac{\bar{\gamma}}{y_1^2+h^2}\right)$ and $\log_2(1+\gamma_E) \to \log_2\left(\frac{\bar{\gamma}}{(x_1-x_2)^2+y_2^2+h^2}\right)$, respectively. Then, the asymptotic SOP can be obtained as

$$\begin{aligned}\mathcal{P}_{so} &\approx \Pr\left(\log_2\left(\frac{(x_1-x_2)^2+y_2^2+h^2}{y_1^2+h^2}\right) \le R_{th}\right) \\ &\approx \Pr\left(\frac{(x_1-x_2)^2+y_2^2+h^2}{y_1^2+h^2} \le C_{th}\right).\end{aligned} \tag{11}$$

The above expression unveils that with a given target secrecy rate R_{th}, the system's SOP will converge to a constant as the transmit power grows, which yields the following corollary.

Corollary 1. *At a high transmit SNR regime, the asymptotic SOP of the considered secure pinching-antenna system can be obtained as a constant value, i.e.,*

$$\mathcal{P}_{so} \approx \frac{2}{D}\int_0^{D/2} F_\chi\left(C_{th}(t^2+h^2) - h^2\right) dt, \tag{12}$$

where $F_\chi(\cdot)$ denotes the CDF of $\chi = (x_1-x_2)^2 + y_2^2$, which can be obtained similarly as the derivation of (5) presented in Appendix A and thus omitted here for brevity.

C. Lower Bound on SOP

Next, to demonstrate the advantage of the pinching antenna in enhancing system's security, we derive a theoretical lower bound on the SOP for the considered pinching-antenna system and compare it with that for traditional FPA systems. Specifically, for FPA systems, we assume that the BS's antenna is deployed at the center of the deployment region size with a height h, i.e., $\hat{\boldsymbol{\psi}}_0 = (0, 0, h)$. As a result, the corresponding SNR at Bob and Eve is given by $\hat{\gamma}_B = \frac{\eta P_s}{\|\hat{\boldsymbol{\psi}}_0 - \boldsymbol{\psi}_1\|^2\sigma^2} = \frac{\bar{\gamma}}{x_1^2+y_1^2+h^2}$ and $\hat{\gamma}_E = \frac{\eta P_s}{\|\hat{\boldsymbol{\psi}}_0 - \boldsymbol{\psi}_2\|^2\sigma^2} = \frac{\bar{\gamma}}{x_2^2+y_2^2+h^2}$, respectively. Hence, the SOP of the conventional FPA system is expressed as $\hat{\mathcal{P}}_{so} = \Pr\left(\log_2\left(1+\hat{\gamma}_B\right) - \log_2\left(1+\hat{\gamma}_E\right) \le R_{th}\right)$.

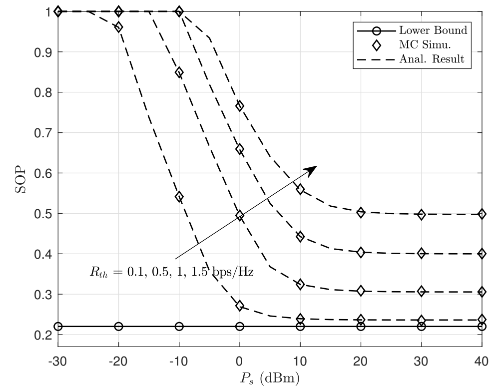

Fig. 2. Secrecy outage probability of the considered secure pinching-antenna system versus the transmit power (in dBm) for different values of R_{th}.

Theorem 2. *For the considered secure pinching-antenna system, the SOP is lower bounded by $\mathcal{P}_{so,low} = \frac{2\pi-1}{24} \approx 0.2201$, while for conventional FPA systems, the SOP is lower bounded by $\hat{\mathcal{P}}_{so,low} = 0.5$.*

Proof: Please refer to Appendix B. □

Theorem 2 demonstrates that regardless of Bob and Eve's distribution region size and BS transmit power, the considered pinching-antenna system can achieve a much lower bound on SOP than its FPA counterpart. This is because for FPA systems, the instantaneous achievable secrecy rate is dominated by the distances between the BS and Bob/Eve due to the fixed antenna layout. Thus, the inherent random distributions of Bob and Eve will result in a high SOP. In contrast, for pinching-antenna systems, the position of the antenna can be flexibly adjusted to reduce the propagation distance between the BS and Bob, thereby enhancing the SOP performance.

IV. Numerical Results

In this section, simulation results are provided to examine the accuracy of our derived analytical results. The simulation parameters are set as follows. The waveguide is deployed at a height of $h = 3$ m. The carrier frequency is set to $f = 28$ GHz and the effective refractive index is $n_{\text{eff}} = 1.4$. Besides, the received noise power is set to $\sigma^2 = -80$ dBm. The number of sampling points for Gaussian-Chebyshev quadrature is set to $N = 100$. In all figures, we present the SOP derived from Monte Carlo (MC) simulations with 10^5 independent channel trials and compare it with our analytical results.

In Fig. 2, we plot the SOP versus the transmit power of the BS based on MC simulations and our analytical expressions in Theorems 1 and 2. The deployment region size is set to $D = 30$ m. As can be observed, the SOP derived from MC simulations matches well with (9) under any value of P_s and R_{th}, which confirms the accuracy of our analytical solution presented in Theorem 1. Besides, we also observe that the SOP in all figures is no less than a constant lower bound, verifying the effectiveness of our analysis in Theorem 2. Furthermore, it can be observed from Fig. 2 that when P_s is small, increasing P_s results in a lower SOP due to different scaling laws of Bob and Eve's SNRs. However, when P_s is large, the SOP cannot

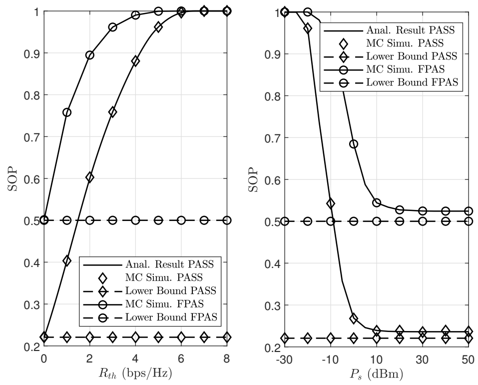

Fig. 3. Performance comparison between pinching-antenna systems (PASS) and conventional fixed-position antenna systems (FPAS).

be further improved with increasing P_s, which is consistent with our analysis in Corollary 1.

In Fig. 3, we compare the SOP of the considered pinching-antenna system with that of conventional FPA systems. The deployment region size is set to $D = 30$ m. Specifically, we plot the SOP versus the target secrecy rate with $P_s = 20$ dBm and the SOP versus the transmit power with $R_{th} = 0.1$ bps/Hz. It can be observed that the pinching-antenna system achieves a better secrecy performance compared to the FPA system. This is attributed to the position flexibility of the pinching antenna to create a stronger line-of-sight link between the BS and Bob. Besides, the results derived from MC simulations align with both of the closed-form SOP presented in Theorem 1 and the lower-bounded SOP presented in Theorem 2.

V. Conclusion

In this paper, we investigated the performance of a secure pinching-antenna system. An approximate expression of the SOP was first derived in closed-form and its asymptotic behavior was analyzed in subsequence. Furthermore, we derived constant lower-bounded SOPs for the considered system and conventional FPA systems, which indicates that the former can be significantly reduced compared to the latter. Finally, our analytical results were verified via simulations, showing that the system's secrecy performance can be significantly improved compared with FPA systems due to the exceptional position flexibility of pinching antennas.

Appendix A
Derivation of Equation (5)

Since $x_1, x_2 \sim \mathcal{U}[-\frac{D}{2}, \frac{D}{2}]$, it is easy to show that $x_1 - x_2$ is a triangular distributed variable over $[-D, D]$. Let us define $X = x_1 - x_2$. The PDF of X^2 can be derived as

$$f_{X^2}(t) = \frac{1}{D\sqrt{t}} - \frac{1}{D^2},\ 0 \le t \le D^2. \tag{13}$$

Then, let $Y = y_2$, with $y_2 \sim \mathcal{U}[-\frac{D}{2}, \frac{D}{2}]$. The PDF of Y^2 can be derived as

$$f_{Y^2}(t) = \frac{1}{D\sqrt{t}},\ 0 \le t \le \frac{D^2}{4}. \tag{14}$$

As a result, the PDF of $W = X + Y$ can be obtained as

$$\begin{aligned} f_W(w) &= \int_A^B f_{X^2}(x) f_{Y^2}(w-x)dx \\ &= \frac{1}{D^2}\int_A^B \frac{1}{\sqrt{x(w-x)}}dx - \frac{1}{D^3}\int_A^B \frac{1}{\sqrt{w-x}}dx \\ &= \frac{1}{D^2} I_1(A,B) - \frac{1}{D^3} I_2(A,B), \end{aligned} \tag{15}$$

where $A = \max(0, w - \frac{D^2}{4})$ and $B = \min(w, D^2)$. Particularly, the two integrals I_1 and I_2 can be calculated as $I_1(A,B) = 2\arcsin\sqrt{\frac{B}{w}} - 2\arcsin\sqrt{\frac{A}{w}}$ and $I_2(A,B) = 2\left(\sqrt{w-A} - \sqrt{w-B}\right)$, respectively [14].

Note that the PDF of $\gamma_E = \frac{\bar{\gamma}}{W+h^2}$ can be obtained as

$$f_{\gamma_E}(z) = \frac{\bar{\gamma}}{z^2} \cdot f_W(\frac{\bar{\gamma}}{z} - h^2). \tag{16}$$

Finally, by substituting (15) into (16) and after some manipulations, we can derive the closed-form of $f_{\gamma_E}(z)$ in (5).

Appendix B
Proof of Theorem 2

For the considered pinching-antenna system, we have

$$\mathcal{P}_{so} \ge \Pr\left(\log_2(1+\gamma_B) - \log_2(1+\gamma_E) \le 0\right), \tag{17}$$

Thus, a lower bound of the SOP can be obtained as

$$\begin{aligned} \mathcal{P}_{so,low} &= \Pr\left(\log_2(1+\gamma_B) - \log_2(1+\gamma_E) \le 0\right) \\ &= \Pr\left(\frac{\bar{\gamma}}{y_1^2 + h^2} \le \frac{\bar{\gamma}}{(x_1-x_2)^2 + y_2^2 + h^2}\right), \end{aligned} \tag{18}$$

which is independent of P_s and R_{th}. Then, let $Z = y_1$, (18) can be further expanded as

$$\begin{aligned} \mathcal{P}_{so,low} =& \Pr\Big(\underbrace{(x_1-x_2)^2}_{X^2} + \underbrace{y_2^2}_{Y^2} \le \underbrace{y_1^2}_{Z^2}\Big) \\ =& 8\int_0^{D/2} \frac{1}{D}dz \int_0^z \frac{D-x}{D^2}dx \int_0^{\sqrt{z^2-x^2}} \frac{1}{D}dy \\ =& \frac{8}{D^3}\int_0^{D/2} dz \underbrace{\int_0^z \sqrt{z^2-x^2}dx}_{J_1} \\ & - \frac{8}{D^4}\int_0^{D/2} dz \underbrace{\int_0^z x\sqrt{z^2-x^2}dx}_{J_2}. \end{aligned} \tag{19}$$

In (19), the integral term J_1 indicates the area of a sector with radius z, for which we have $J_1 = \frac{\pi z^2}{4}$. Besides, defining $u = z^2 - x^2$, the integral term J_2 can be calculated as $J_2 = \int_{u=z^2}^{u=0} -\frac{\sqrt{u}}{2}du = \frac{1}{2}\int_0^{z^2}\sqrt{u}du = \frac{z^3}{3}$.

By substituting J_1 and J_2 into (19), we have

$$\begin{aligned} \mathcal{P}_{so,low} &= \frac{8}{D^3}\int_0^{D/2} \frac{\pi z^2}{4}dz - \frac{8}{D^4}\int_0^{D/2} \frac{z^3}{3}dz \\ &= \frac{2\pi}{D^3} \cdot \frac{(D/2)^3}{3} - \frac{8}{3D^4} \cdot \frac{(D/2)^4}{4} = \frac{2\pi - 1}{24}. \end{aligned} \tag{20}$$

For conventional FPA systems, we can derive the lower bound on its SOP, i.e., $\hat{\mathcal{P}}_{so,low} = 0.5$, in the same way.

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